Cisco CCNP

4 OSPF Variants

Written By
Cambell Xiao

Purpose:

Configuring Multi Area OSPF using 4 different areas with stubby, totally stubby, not so stubby, and normal OSPF. Ensure connectivity between all routers and an external route through the NSSA area. Finding and capturing every Link State Advertisement (explained in background info) that is sent between each area with an OSPF variant. Capturing these link state advertisements will be done on wireshark.

Background Info

Even though running normal OSPF is perfectly fine on most networks, in some special cases, it is more beneficial to run an OSPF variant either to maximize the router's resources, easier management of the network, and security. All OSPF instances run through the process of sharing Link State Advertisements with neighboring routers to learn of how to reach them. Link State Advertisements 1-3 share the location of all other OSPF routers that are connected to the OSPF Area and LSA 4-5 share the location of any ASBR, routers connected to an external routing protocol, into the OSPF area. When a router receives these advertisements, they will insert the routes to these networks into its routing table and Link State Database (a list of all nodes that run OSPF and how to reach them which is shared among all routers in the OSPF area). This process can be very CPU intensive due to the size of the Link State Database in topologies with many different routers and therefore many different routes. All OSPF instances run depending on what area they are in. OSPF areas are like routers grouping together, usually based on physical location. Many OSPF Areas make up one larger OSPF instance. By separating these routers, type 1 and 3 LSAs will only be routed overhead in their own areas and not the whole network. OSPF variants such as Stubby, Totally Stubby, and Not So Stubby will have different interactions with these link state advertisements. Stubby and Totally Stubby areas run extremely similarly. Both areas will only originate type 1-3 LSAs in these areas, meaning that they will still advertise their network to other OSPF areas. However, Stubby and Totally Stubby areas will not insert routes into their own routing table, leaving the only way to exit their area with a default route. As such, these areas cannot locate the best path to get to a destination, as they will always just forward packets to the nearest ABR regardless of whether it's the most efficient way to reach the other network. Stubby OSPF will have a smaller Link State Database and a smaller routing table as a result and will therefore use less CPU and Memory. The main difference between Totally Stubby and Stub networks is that Totally Stubby networks do

not have internal routes, so they cannot reach other routers in their own area, other than the directly connected routers. Totally stubby networks are used to completely minimize the CPU and memory drain of running OSPF on routers, while Stub networks are used to generally lower the CPU and memory drain whilst still maintaining

connectivity in the area. In our topology, a backbone area is needed to connect the NSSA, Stub, and Totally Stub networks because only the backbone area can originate Type 3 LSAs and is needed to transmit information between the Stub networks. If two stub networks are directly connected, they will be stuck in a routing loop where one router will have a default route to the other router and will be suck transmitting packets between them. Type 1-2 LSAs are sent by the DR of each OSPF area. These LSAs are flooded in the area to allow connectivity between each router in the area this is where inter area ospf routes are made. Type 3 LSAs are sent by the ABR towards the backbone area to allow Multi Area OSPF to function by informing routers outside their area the routes inside their network. Type 4 LSAs are only generated in Multi Area OSPF configurations, they advertise the ip and prefix of any ABRs and ASBRS. Type 5 LSAs are only generated by ASBRs when a default-information originate command is sent out. This LSA is sent only to non-stubby areas and does not contain the necessary information to reach the ASBR, so all ABRs that receive these advertisements send out a type 4 LSA which contains the ASBR's location.Type 7 LSAs are sent by Not So Stubby Areas, when a no-redistribution and a default-information-originate command is sent. This will flood the NSSA area with type 7 LSAs which is a route to the outside network and all ABRs that receive this will send out a type 5 LSA so routers know where to send packets if they want to reach an outside network.

Lab Summary

To configure all OSPF variants properly, we created a backbone area that would run normal OSPF in order to allow routing of Link State advertisements properly to all other networks. First, we configured Multi Area OSPF on all routers. Configuring the stub network, we created a default route out towards the backbone area as stubby areas will only have default routes towards ABRs. Second, by doing Area 1 stub, we configured the area to be a stub network which would immediately delete all routes in the routing table and Link State Database to all other OSPF routers. The ABR would then inject a default route into the stub router to allow connectivity outside. Next, we configured the Totally Stubby routers. These routers would follow the same process as configuring stub

routers, except we had to configure the router with an area 2 stub no-summary to block all type 3 LSAs from entering the network. This way, the Totally Stubby Routers will have no OSPF routes to outside networks and have no OSPF routes to other routes in the network to maximize the saving of memory and CPU. Configuring the NSSA was the most difficult task. We had to configure the router using the command “Area 3 NSSA” in order to force the network to become an NSSA area, then we configured an external router directly connected to the NSSA that is not configured with OSPF. Then, we set up a default route on the NSSA which directly leads to the external router and used the command “area 3 NSSA no-redistribution default-information-originate". This command will cause the NSSA to broadcast type 7 LSAs only in the NSSA’s area. The ABR of the area will then translate these type 7 LSAs into type 5 LSAs to broadcast. Using wireshark, we collected type 1-3 LSAs from the backbone area, type 4 and 5 from a stubby network that is configured with normal OSPF due to stubby networks denying all type 5 LSAs, and a type 7 from the NSSA’s network.

Lab Commands

Router ospf [process id]

> Creates an OSPF instance on the router, enables OSPF routing and also enters routing configuration mode to allow setting up of ipv4 OSPF.

Network [network address] [wildcard mask] area [area number]

> Used to tell other OSPF routers how to reach your directly connected networks. Also used to configure what area your directly connected area is. This command is only used for IPv4 connectivity.

Area [area] stub

Configures the area as a stub area.

Area [area] nssa

Configures the area as a not so stubby area.

Area [area] nssa no-redistribution default-information-originate

Forces the propagation of any default routes made on the router. This is only used on ASBRs because it distributes type 7 LSAs into the area. Type 5 LSAs will be redistributed through all other ABRs in other areas.

Router-Id

Configures a manually set Router id to influence OSPF election process.

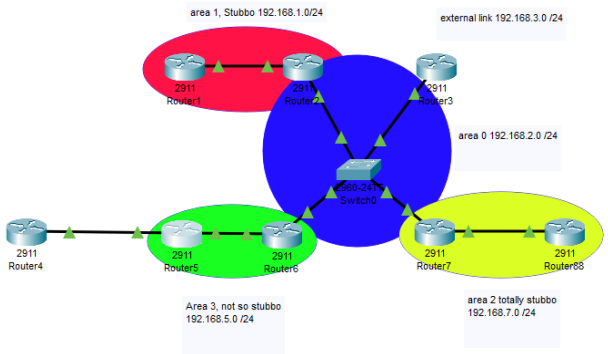
Area [area] stub no-summary

Turns the area into a totally stubby area by forcefully turning off all summary advertisements in the area, so totally stubby routers do not have an ospf route to each other.

Monitor session [session number] [source/destination] interface [interface number]

This allowed our switch to forward all LSAs and OSPF hello packets into our PC which has Wireshark open. Without this command, routers would not route their LSAs and OSPF packets into an end device because it's a security risk and is just completely useless.

Network Diagram




```
O*IA   0.0.0.0/0 [110/2] via 192.168.1.2, 00:24:32,
GigabitEthernet0/0/1

      192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks

C        192.168.1.0/24 is directly connected,
GigabitEthernet0/0/1
```

```
L        192.168.1.1/32 is directly connected,
GigabitEthernet0/0/1

O IA  192.168.2.0/24 [110/2] via 192.168.1.2, 00:24:27,
GigabitEthernet0/0/1

O IA  192.168.5.0/24 [110/3] via 192.168.1.2, 00:24:17,
GigabitEthernet0/0/1

O IA  192.168.7.0/24 [110/3] via 192.168.1.2, 00:24:17,
GigabitEthernet0/0/1

      192.168.8.0/24 is variably subnetted, 2 subnets, 2 masks

C        192.168.8.0/24 is directly connected, Loopback0

L        192.168.8.1/32 is directly connected, Loopback0

      192.168.9.0/32 is subnetted, 1 subnets

O IA     192.168.9.1 [110/4] via 192.168.1.2, 00:24:17,
GigabitEthernet0/0/1

      192.168.10.0/32 is subnetted, 1 subnets

O IA     192.168.10.1 [110/4] via 192.168.1.2, 00:24:17,
GigabitEthernet0/0/1
```

R2

```
S*    0.0.0.0/0 is directly connected, GigabitEthernet0/0/0

      192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks

C        192.168.1.0/24 is directly connected,
GigabitEthernet0/0/1

L        192.168.1.2/32 is directly connected,
GigabitEthernet0/0/1

      192.168.2.0/24 is variably subnetted, 2 subnets, 2 masks

C        192.168.2.0/24 is directly connected,
GigabitEthernet0/0/0

L        192.168.2.1/32 is directly connected,
GigabitEthernet0/0/0

O IA  192.168.5.0/24 [110/2] via 192.168.2.3, 00:31:36,
GigabitEthernet0/0/0

O IA  192.168.7.0/24 [110/2] via 192.168.2.4, 00:31:36,
GigabitEthernet0/0/0
```

```
      192.168.8.0/32 is subnetted, 1 subnets
O        192.168.8.1 [110/2] via 192.168.1.1, 00:31:46,
GigabitEthernet0/0/1
      192.168.9.0/32 is subnetted, 1 subnets
O IA     192.168.9.1 [110/3] via 192.168.2.3, 00:31:36,
GigabitEthernet0/0/0
      192.168.10.0/32 is subnetted, 1 subnets
O IA     192.168.10.1 [110/3] via 192.168.2.4, 00:31:36,
GigabitEthernet0/0/0
```

R3

```
O*E2  0.0.0.0/0 [110/1] via 192.168.2.3, 00:42:50,
GigabitEthernet0/0/0
O IA  192.168.1.0/24 [110/2] via 192.168.2.1, 00:34:20,
GigabitEthernet0/0/0
      192.168.2.0/24 is variably subnetted, 2 subnets, 2 masks
C        192.168.2.0/24 is directly connected,
GigabitEthernet0/0/0
L        192.168.2.2/32 is directly connected,
GigabitEthernet0/0/0
O IA  192.168.5.0/24 [110/2] via 192.168.2.3, 00:53:37,
GigabitEthernet0/0/0
O IA  192.168.7.0/24 [110/2] via 192.168.2.4, 00:36:54,
GigabitEthernet0/0/0
      192.168.8.0/32 is subnetted, 1 subnets
O IA     192.168.8.1 [110/3] via 192.168.2.1, 00:34:20,
GigabitEthernet0/0/0
      192.168.9.0/32 is subnetted, 1 subnets
O IA     192.168.9.1 [110/3] via 192.168.2.3, 00:53:29,
GigabitEthernet0/0/0
      192.168.10.0/32 is subnetted, 1 subnets
O IA     192.168.10.1 [110/3] via 192.168.2.4, 00:36:54,
GigabitEthernet0/0/0
```

R4

```
      192.168.20.0/24 is variably subnetted, 2 subnets, 2 masks
C        192.168.20.0/24 is directly connected,
GigabitEthernet0/0/0
L        192.168.20.1/32 is directly connected,
GigabitEthernet0/0/0
```

R5

```
S*    0.0.0.0/0 is directly connected, GigabitEthernet0/0/0
O IA  192.168.1.0/24 [110/3] via 192.168.5.2, 00:35:30,
GigabitEthernet0/0/1
O IA  192.168.2.0/24 [110/2] via 192.168.5.2, 00:35:44,
GigabitEthernet0/0/1
      192.168.5.0/24 is variably subnetted, 2 subnets, 2 masks
C        192.168.5.0/24 is directly connected,
GigabitEthernet0/0/1
L        192.168.5.1/32 is directly connected,
GigabitEthernet0/0/1
O IA  192.168.7.0/24 [110/3] via 192.168.5.2, 00:35:44,
GigabitEthernet0/0/1
      192.168.8.0/32 is subnetted, 1 subnets
O IA     192.168.8.1 [110/4] via 192.168.5.2, 00:35:30,
GigabitEthernet0/0/1
      192.168.9.0/24 is variably subnetted, 2 subnets, 2 masks
C        192.168.9.0/24 is directly connected, Loopback0
L        192.168.9.1/32 is directly connected, Loopback0
      192.168.10.0/32 is subnetted, 1 subnets
O IA     192.168.10.1 [110/4] via 192.168.5.2, 00:35:44,
GigabitEthernet0/0/1
      192.168.20.0/24 is variably subnetted, 2 subnets, 2 masks
C        192.168.20.0/24 is directly connected,
GigabitEthernet0/0/0
L        192.168.20.2/32 is directly connected,
GigabitEthernet0/0/0
```

R6

```
O*N2  0.0.0.0/0 [110/1] via 192.168.5.1, 00:44:22,
GigabitEthernet0/0/1

O IA  192.168.1.0/24 [110/2] via 192.168.2.1, 00:35:52,
GigabitEthernet0/0/0

      192.168.2.0/24 is variably subnetted, 2 subnets, 2 masks

C        192.168.2.0/24 is directly connected,
GigabitEthernet0/0/0

L        192.168.2.3/32 is directly connected,
GigabitEthernet0/0/0

      192.168.5.0/24 is variably subnetted, 2 subnets, 2 masks

C        192.168.5.0/24 is directly connected,
GigabitEthernet0/0/1

L        192.168.5.2/32 is directly connected,
GigabitEthernet0/0/1

O IA  192.168.7.0/24 [110/2] via 192.168.2.4, 00:36:06,
GigabitEthernet0/0/0

      192.168.8.0/32 is subnetted, 1 subnets

O IA     192.168.8.1 [110/3] via 192.168.2.1, 00:35:52,
GigabitEthernet0/0/0

      192.168.9.0/32 is subnetted, 1 subnets

O        192.168.9.1 [110/2] via 192.168.5.1, 00:55:00,
GigabitEthernet0/0/1

      192.168.10.0/32 is subnetted, 1 subnets

O IA     192.168.10.1 [110/3] via 192.168.2.4, 00:36:06,
GigabitEthernet0/0/0
```

R7

```
S*    0.0.0.0/0 is directly connected, GigabitEthernet0/0/0

O IA  192.168.1.0/24 [110/2] via 192.168.2.1, 00:36:29,
GigabitEthernet0/0/0
```

```
      192.168.2.0/24 is variably subnetted, 2 subnets, 2 masks
C        192.168.2.0/24 is directly connected,
GigabitEthernet0/0/0
L        192.168.2.4/32 is directly connected,
GigabitEthernet0/0/0
O IA  192.168.5.0/24 [110/2] via 192.168.2.3, 00:39:03,
GigabitEthernet0/0/0
      192.168.7.0/24 is variably subnetted, 2 subnets, 2 masks
C        192.168.7.0/24 is directly connected,
GigabitEthernet0/0/1
L        192.168.7.2/32 is directly connected,
GigabitEthernet0/0/1
      192.168.8.0/32 is subnetted, 1 subnets
O IA     192.168.8.1 [110/3] via 192.168.2.1, 00:36:29,
GigabitEthernet0/0/0
      192.168.9.0/32 is subnetted, 1 subnets
O IA     192.168.9.1 [110/3] via 192.168.2.3, 00:39:03,
GigabitEthernet0/0/0
      192.168.10.0/32 is subnetted, 1 subnets
O        192.168.10.1 [110/2] via 192.168.7.3, 00:39:03,
GigabitEthernet0/0/1
```

R8

```
O*IA  0.0.0.0/0 [110/2] via 192.168.7.2, 00:39:30,
GigabitEthernet0/0/1
      192.168.7.0/24 is variably subnetted, 2 subnets, 2 masks
C        192.168.7.0/24 is directly connected,
GigabitEthernet0/0/1
L        192.168.7.3/32 is directly connected,
GigabitEthernet0/0/1
      192.168.10.0/24 is variably subnetted, 2 subnets, 2 masks
C        192.168.10.0/24 is directly connected, Loopback0
L        192.168.10.1/32 is directly connected, Loopback0
```

Configurations

R1:

```
hostname r1
int g0/0/1
ip add 192.168.1.1 255.255.255.0
no shut
int lo0
ip add 192.168.8.1 255.255.255.0

router ospf 10
network 192.168.1.0 0.0.0.255 area 1
network 192.168.8.0 0.0.0.255 area 1
area 1 stub
router-id 1.1.1.1
```

R2:

```
 hostname R2
int g0/0/0
ip add 192.168.2.1 255.255.255.0
no shut
exit

int g0/0/1
ip add 192.168.1.2 255.255.255.0
no shut
router ospf 10
```

```
network 192.168.1.0 0.0.0.255 area 1
network 192.168.2.0 0.0.0.255 area 0
area 1 stub
router-id 2.2.2.2

ip route 0.0.0.0 0.0.0.0 g0/0/0
```

R3

```
hostname r3
int g0/0/0
ip add 192.168.2.2 255.255.255.0
no shut

int g0/0/1
ip add 192.168.3.2 255.255.255.0
no shut

router ospf 10
network 192.168.2.0 0.0.0.255 area 0
router-id 3.3.3.3
```

R4

```
hostname r4

int g0/0/0
ip add 192.168.20.1 255.255.255.0
no shut
router eigrp 10
```

```
network 192.168.20.0
---------------------------------------------
R5
hostname r5
int g0/0/1
ip add 192.168.5.1 255.255.255.0
no shut

int lo0
ip add 192.168.9.1 255.255.255.0

int g0/0/0
ip add 192.168.20.2 255.255.255.0
no shut

router ospf 10
network 192.168.5.0 0.0.0.255 area 3
network 192.168.9.0 0.0.0.255 area 3
area 3 nssa
router-id 5.5.5.5
area 3 nssa no-redistribution default-information-originate
---------------------------------------------
R6
en
conf t
hostname r6

int g0/0/0
```

```
ip add 192.168.2.3 255.255.255.0
no shut

int g0/0/1
ip add 192.168.5.2 255.255.255.0
no shut

router ospf 10
network 192.168.5.0 0.0.0.255 area 3
network 192.168.2.0 0.0.0.255 area 0
area 3 nssa
router-id 6.6.6.6
```

```
R7
en
conf t
hostname R7
int g0/0/0
ip add 192.168.2.4 255.255.255.0
no shut

int g0/0/1
ip add 192.168.7.2 255.255.255.0
no shut

ip route 0.0.0.0 0.0.0.0 g0/0/0

router ospf 10
```

```
network 192.168.2.0 0.0.0.255 area 0
network 192.168.7.0 0.0.0.255 area 2
area 2 stub no-summary
router-id 7.7.7.7
----------------------------------------
R8
en
conf t
hostname R8

int g0/0/1
ip add 192.168.7.3 255.255.255.0
no shut

int lo0
ip add 192.168.10.1 255.255.255.0

router ospf 10
network 192.168.7.0 0.0.0.255 area 2
network 192.168.10.0 0.0.0.255 area 2
area 2 stub no-summary
router-id 8.8.8.8
```

Problems

Unfamiliar commands

When setting up the NSSA, we did a normal default-information-originate instead of a no-redistribution default-information-originate. This caused the NSSA to send out type 5 LSAs instead of type 7 LSAs.

Unfamiliarity with how Stub networks worked

We were not aware that Stub networks did not require a default-information-originate. And when we did set up a default-information-originate, the network broke. We did ask other groups about this issue but nobody else was familiar with it as well, one group stated that they used static routes to connect the OSPF routers.

Faulty cables

Sometimes when we were configuring routers with ip addresses, the interfaces would not turn on. This was due to a faulty cable which was actually a lot more common than we thought it would be.

Switch not configured as port monitoring.

We were unable to receive any LSAs or OSPF hello packets at all until we set up port monitoring. This problem took us a day or two to fix until the instructor gave us a hint.

Connecting a pc between two routers.

Due to a router only having two ports, we needed to use a switch to ensure connectivity between the routers whilst still being able to receive LSA packets after connecting our PC to the switch.

Conclusion

OSPF is a very versatile routing protocol which may seem cost effective at a glance, but its many OSPF Variants can make up for this issue. Stubby and Totally stubby OSPF variants are used to mitigate the effects of a larger OSPF topology. By decreasing the CPU and memory usage of these OSPF configured routers, the scalability of an OSPF network increases greatly because the only reason OSPF isn't very scalable, is the CPU and memory cost of saving such a large link state database in each router. An NSSA is used to ensure connectivity with routers that may run on a different routing protocol whilst still ensuring that the cost of CPU and memory is also minimal. It does keep inter

area routes (routes to other routers in your area) but the only connectivity outside your area is with a default route. While these OSPF setups may seem complicated to set up, they are highly efficient in ensuring a scalable topology whilst maintaining connectivity within networks.

SIGN OFF

Types of OSPF sign off

Cambell Xiao P3-4 CCNP Mr. Mason